



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

KARAN DAVE, 26070122507, CSE B3, 2025-2029

ASSIGNMENT No: 11

TITLE: Write a Java program to demonstrate the implementation of interfaces.

Problem Statement

Write a Java program to demonstrate the implementation of interfaces.

Learning Objectives

To implement interfaces for achieving abstraction and multiple inheritance of type.

Theory

An interface specifies a set of abstract methods that implementing classes must define. It provides complete abstraction and supports multiple inheritance of type. Interfaces improve modularity, loose coupling, and software flexibility. They allow different classes to implement common behavior while maintaining independent implementations. Interfaces are widely used in enterprise application development.



Exercise

1. Create an interface Printable and implement it in Student and Employee classes.

```
1 interface Printable {
2     void printDetails();
3 }
4 class Student implements Printable {
5     String name;
6     int rollNo;
7     Student(String name, int rollNo) {
8         this.name = name;
9         this.rollNo = rollNo;
10    }
11    public void printDetails() {
12        System.out.println("Student Name: " + name);
13        System.out.println("Roll No: " + rollNo);
14    }
15 }
16 class Employee implements Printable {
17     String name;
18     int employeeId;
19 }
20 Employee(String name, int employeeId) {
21     this.name = name;
22     this.employeeId = employeeId;
23 }
24 public void printDetails() {
25     System.out.println("Employee Name: " + name);
26     System.out.println("Employee ID: " + employeeId);
27 }
28 public class Main {
29     public static void main(String[] args) {
30         Student student = new Student("Karan", 101);
31         Employee employee = new Employee("Sujay", 501);
32         student.printDetails();
33         System.out.println();
34         employee.printDetails();
35     }
36 }
```

Problems @ Javadoc Declaration Console Properties
<terminated> Main (2) [Java Application] C:\Users\pdave\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64_21.0.11.v20260515-1531\jre\bin\javaw.exe (16-Aug-2026, 11:49:58)
Student Name: Karan
Roll No: 101

Employee Name: Sujay
Employee ID: 501

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Student Name: Karan

Roll No: 101

Employee Name: Sujay

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2. Create an interface Switchable with a method turnOn(). Implement the interface in Light and Fan classes to display the device status.

```
1 interface Switchable {
2     void turnOn();
3 }
4 class Light implements Switchable {
5     public void turnOn() {
6         System.out.println("Light is ON");
7     }
8 }
9 class Fan implements Switchable {
10    public void turnOn() {
11        System.out.println("Fan is ON");
12    }
13 }
14 public class Program2 {
15     public static void main(String[] args) {
16         Light light = new Light();
17         Fan fan = new Fan();
18         light.turnOn();
19         fan.turnOn();
20     }
21 }
```

Problems @ Javadoc Declaration Console Properties
<terminated> Program2 [Java Application] C:\Users\pdave\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64_21.0.11.v20260515-1531\jre\bin\javaw.exe (16-Aug-2026, 11:53:30)
Light is ON
Fan is ON

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Light is ON
Fan is ON